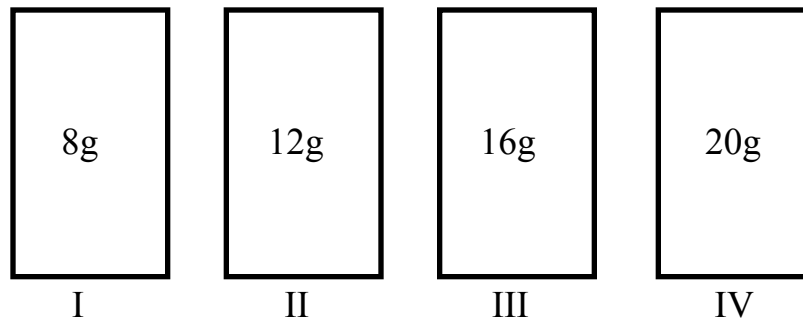

Experiment: Patch Assessment Test

File name: patch_assess.csv

Design:



Duration: 4 days

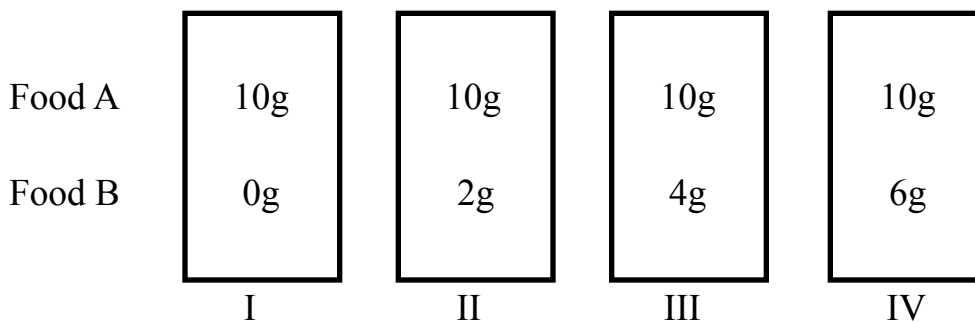
Food type: Oxalate and untreated (food types alternated at each station on each day)

Preliminary analysis: (1) increase in GUDs with increasing initial density (i.e., not equalizing GUDs), (2) lower GUDs for untreated seeds than oxalate seeds, (3) interaction between initial density and food type such that rabbits better equalized GUDs for untreated seeds than for oxalate seeds

Experiment: Apparent Competition Test

File name: AC_oat_back.csv and AC_sun_back.csv

Design:



Duration: 8 days

Food type: Oats and sunflower (2 experiments with data analyzed separately; first 4 days consisted of oats as Food A and sunflower as Food B and vice versa on next 4 days)

Preliminary analysis: (1) negative effect of initial sunflower density on oat GUDs, (2) negative effect of initial oat density on sunflower GUDs

Comment: After finding (-,0) effects in my AR/SD tests (see below), I was concerned that rabbits were not foraging in a density-dependent manner, which was the motivation for the patch assessment and apparent competition experiments.

Experiment: Associational Refuge/Shared Doom Test

File name: AR.csv and SD.csv

Design:

Food A	10g	10g	10g	10g
Food B	0g	2g	4g	6g
	I	II	III	IV

Duration: 8 days

Food type: Oxalate and untreated (2 experiments with data analyzed separately; first 4 days consisted of untreated seeds as Food A and oxalate seeds as Food B and vice versa on next 4 days)

Preliminary analysis: (1) no effect of initial oxalate density on untreated GUDs, (2) negative effect of initial untreated density on oxalate GUDs

Experiment: Associational Refuge Test (with oxalate supplement)

File name: AR_ox_supp.csv

Design:

Untreated	10g	10g	10g	10g
Oxalate	0g	2g	4g	6g
	I	II	III	IV

(12g) Oxalate

Duration: 4 days

Food type: Oxalate and untreated (supplemental oxalate seeds were provided in a dish w/o sand)

Preliminary analysis: (weak) positive effect of initial oxalate density on untreated GUDs

Experiment: Scale Test

File name: scale.csv

Design:

Untreated	10g	0g	10g	10g
Oxalate	0g	10g	10g	10g
	I	II	III	IV

Duration: 4 days

Food type: Oxalate and untreated

Preliminary analysis: (1) [weak] positive effect of oxalate on untreated when both types are thoroughly mixed (I vs III), (2) no effect of oxalate on untreated when each type occupies one half of the patch (I vs IV), (3) negative effect of untreated on oxalate when both types are mixed (II vs III), (4) “positive” effect of untreated on oxalate when each type occupies one half of the patch (II vs. IV)

Comment: The preliminary analysis is based on actual densities (g/L) and not just the mass of seeds remaining. At first blush, #4 seems like a peculiar result. However, when you consider that the rabbits seem to be able to recognize the separation of food types in patch IV, the result is consistent with the results for oxalate seeds in the patch assessment experiment; i.e., the rabbits exhibit density-independent foraging on oxalate seeds. In the scale experiment, the initial density of oxalate seeds in patch IV is twice that of the oxalate seeds in patch II, so you would expect the oxalate GUD in IV to be twice as high as the oxalate GUD in II (and it roughly is).

Experiment: Replacement Series

File name: replacement.csv

Design:

Untreated	16g	12g	8g	4g	0g
Oxalate	0g	4g	8g	12g	16g
	I	II	III	IV	V

Duration: 5 days

Food type: Oxalate and untreated

Preliminary analysis: (1) *proportion* of untreated seeds consumed decreases with increasing initial density of oxalate seeds, (2) *proportion* of oxalate seeds consumed increases with increasing initial density of untreated seeds

Comment: The design of this experiment does not follow from patch use theory. And the result is trivial considering that rabbits do not forage selectively and harvest untreated seeds to much lower GUDs than oxalate seeds. However, it seems that many of the papers on AR from the plant-animal interaction literature don't make the critical distinction between addition and replacement experiments. The salient point is that you would conclude that there is strong AR in this system from the replacement experiments, but the addition experiments suggest that AR is much weaker.

Experiment: Composition by Density (2x2 factorial)

File name: comp_density.csv

Design:

Untreated	6g	2g	12g	4g
Oxalate	2g	6g	4g	12g
	I	II	III	IV

Duration: 4 days

Food type: Oxalate and untreated

Preliminary analysis: (1) when 75% of seeds in patch are oxalate, rabbits consume same *proportion* of untreated (and oxalate) seeds from low and high density patches (II v IV), (2) when 25% of seeds in patch are oxalate, rabbits consume greater *proportion* of untreated (and oxalate) seeds from high density patches than low density patches (I v III)

Comment: The patch assessment experiment shows that, in patches with only oxalate seeds, rabbits are density-independent foragers and, in patches with only untreated seeds, rabbits are density-dependent foragers. This experiment was a first attempt to see if rabbits exhibited similar patterns when both food types were in a single patch. Indeed, the rabbits exhibit density-dependent foraging when patches are comprised primarily of untreated seeds and density-independent foraging when patches are primarily comprised of oxalate seeds.